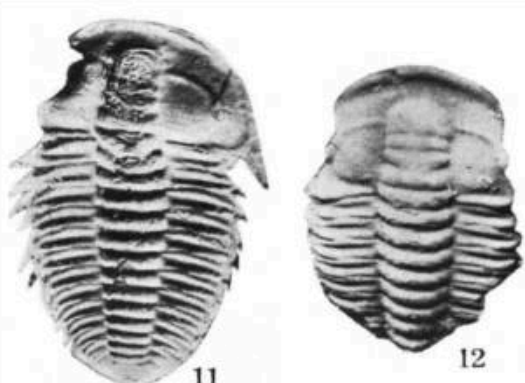


Palaeolenus lantenoisi

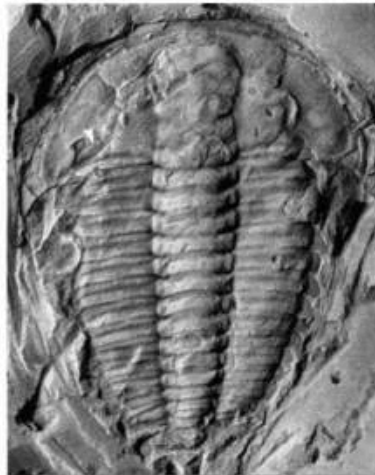
Order/Family	Order Redlichiida / Suborder Redlichiina / Superfamily Ellipsocephaloidea / Family Palaeolenidae		
Genus	Genus Palaeolenus Mansuy, 1912	Author	Mansuy H.
Etymology	palae, Greek, ancient; Olenus, a trilobite genus name		
Species	P. lantenoisi Mansuy, 1912	Author	Mansuy H.
Etymology	Lantenois, a surname		
Age	Cambrian, Series 2, Stage 4	Main distribution	Yunnan, Guizhou, Sichuan, Shaanxi, Hubei
Synonyms	Palaeolenus kunmingensis; P. tingi; P. minor; P. planilimbatus Megapalaeolenus nanzhengensis; M. quadratus; P. yangziensis; P. liantuoensis; M. xixiangensis; M. obsoletus; M. yichangensis		
Diagnosis Characters	Body relatively small, generally about 1 cm long, dorsal shield elongate-oval. Cephalon semicircular; Cranidium subsquare, wider than long; Glabella rectangular, flat-convex, sides subparallel, with four pairs of glabellar furrows, the first pair short and shallow; occipital furrow straight, occipital ring widest medially; fixed cheek relatively broad; eye ridge rather long, palpebral lobe rather small; inner border wide, outer border narrow. Anterior facial suture branches curving slightly forward and outward Obliquely outward, posterior branch short; Free cheek relatively narrow, genal spines short and small, slightly obliquely backward and outward. Thorax of 13 segments, axis slightly narrower than pleural lobes, axis smooth or with small nodes, pleural spines short. Pygidium small, divided into two segments, the anterior one narrow.		
Remarks	/		
 Early illustrated specimens Yunnan; Lu Yanhao et al. 1965  Reproduced from Qian et al. 1985  Complete dorsal shield (Yunnan; He Ming)			
 Complete dorsal shield (Yunnan; Baishui)  Complete dorsal shields on a slab (Yunnan; Guodong) Complete dorsal shield (Yunnan; Wuxing) 			

Palaeolenus douvillei

Order/Family	Order Redlichiida / Suborder Redlichiina / Superfamily Ellipsocephaloidea / Family Palaeolenidae		
Genus	Genus Palaeolenus Mansuy, 1912		Author Mansuy H.
	Etymology palae, Greek, ancient; Olenus, a trilobite genus name		
Species	P. douvillei Mansuy, 1912		Author Mansuy H.
	Etymology Douville, a surname		
Age	Cambrian, Series 2, Stage 4	Main distribution	Yunnan, Guizhou, Sichuan, Shaanxi, Hubei
Synonyms	Palaeolenus xingjiangensis; P. deprati; P. brevicus; Megapalaeolenus deprati; M. guizhouensis; M. expansus; M. magus; M. luquanensis; M. majiashanensis; M. granulatus		
Diagnosis Characters	Dorsal shield elongate-oval, generally somewhat larger than in P. lantenoisi. Cephalon semicircular; cranium subquadrate, wider than long; glabella distinctly convex, slightly expanded anteriorly and slightly constricted at midlength; with four pairs of glabellar furrows; occipital furrow narrow and deep laterally, shallow and broad medially; occipital ring widest medially, its posterior margin sometimes bearing a small occipital node; eye ridge rather long; palpebral lobe of moderate size; fixed cheek broad, posterior limb relatively short; inner and outer borders of nearly equal width; outer border convex; free cheek of moderate width, genal spine rather long; anterior facial suture branches diverging outward, posterior branches directed outward and backward. Thor ax of 14-15 segments, axis smooth or with median nodes; pleural lobes slightly wider than the axis, pleural spines short. Pygidium small, semicircular-oval; divided into two segments; border narrow and flat.		
Remarks	This species is the type species of Palaeolenus.		



Type specimen (Yunnan; Lin Tianrui et al., 2004)



The unique type of the former "M. deprati"
(left side of cephalon deformed; Yunnan; Lin Tianrui et al., 2004)



Figured specimen (Hu Shixue et al., 2013)



Complete dorsal shield (Yunnan; Mask)



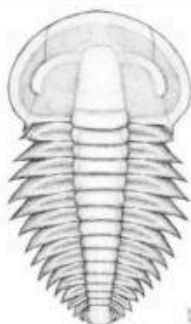
Complete dorsal shield (Yunnan; Guodong)

Yunnanocephalus yunnanensis

Order/Family	Order Redlichiida / Suborder Redlichiina / Superfamily Ellipsocephaloidea / Family Yunnanocephalidae		
Genus	Genus Yunnanocephalus Kobayashi, 1936		Author Kobayashi T.
	Etymology Yunnan; -cephalus, Latin, head		
Species	Y. yunnanensis (Mansuy, 1912)		Author Mansuy H.
	Etymology Yunnan; type specimen type locality		
Age	Cambrian, Series 2, Stage 3	Main distribution	Yunnan
Synonyms	Ptychoparia yunnanensis; Pseudoptychoparia yunnanensis; Yunnancephalus kunyangensis; Yunnancephalus latus Yunnancephalus planifrons; Yunnancephalus subparallelus		
Diagnosis Characters	Dorsal shield elongate-oval, broadly rounded anteriorly and pointed posteriorly. Cephalon elongate-oval; cranium subquadrate; glabella truncate-conical, with three pairs of shallow glabellar furrows; inner border broad and concave, outer border convex; eye ridges stout and long, arising from the sides of the frontal lobe; palpebral lobes convex, small to medium; fixed cheek relatively broad, with an extremely short spine at its posterolateral corner; anterior and posterior facial suture branches both directed forward or Extending backward; Free cheek small and narrow, without genal spine. Thorax of 14 segments; Each ring with axial node or spine, axial spines especially distinct on posterior several rings; 4th or 5th pleural segment longest; Pleural furrow shallow and broad, pleural spines short and stout. Pygidium small, consisting of two axial rings, the anterior ring narrow, the posterior one extending sideways as a broad and stout structure. During the meraspid period the sutures are proparian, becoming gonatoparian in adults.		
Remarks	This species is the type species of Yunnanocephalus.		



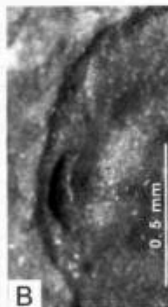
Type specimen (Yunnan; Peng Shanchi 2020)



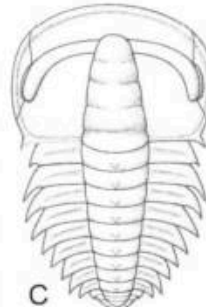
Reconstruction (Shu et al., 1995)



A



B



C

Meraspid stage proparian facial suture specimens and reconstruction Dai et al., 2008



(Yunnan; Qianli zhi Jiang)



(Yunnan; Gao Zishuo)
Complete dorsal shield
(Yunnan; Zai Lushang)
Specimen with axial spine
preserved Yunnan; Lixia

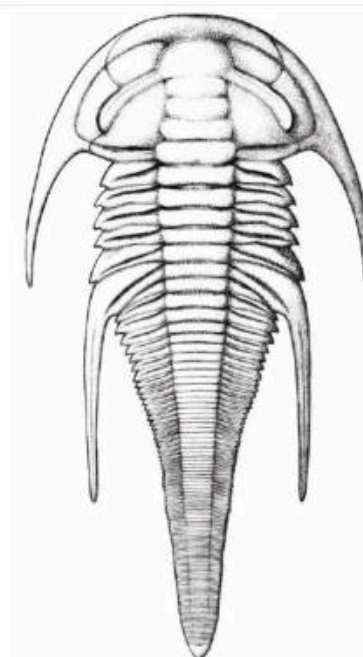


Balcoracania dailyi

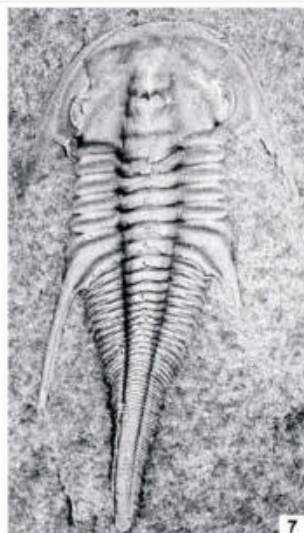
Order/Family	Order Redlichiida / Suborder Redlichiina / Superfamily Emuelloidea / Family Emuellidae		
Genus	Genus Balcoracania Pocock, 1970		Author Pocock K. J.
	Etymology Balcoracania Creek, in the Flinders Range, Australia		
Species	B. dailyi Pocock, 1970		Author Pocock K. J.
	Etymology Daily — named for Dr. B. Daily, the first discoverer of balcoracaniid trilobites		
Age	Cambrian, Series 2, Stage 3	Main distribution	Australia
Synonyms	Balcoracania flindersi		
Diagnosis	Cephalon semicircular; cranidium nearly square; glabella constricted at midlength with an expanded frontal lobe, three pairs of transversely broad glabellar furrows; palpebral lobe nearly as long as the eye ridge, both stout; fixed cheek broad; free cheek narrow; Genal spines of moderate length. The number of thoracic segments may have no fixed limit, currently recorded as 103 (6+97), prothorax and opisththorax divided by the macropleural spine of the 6th segment. Pygidium very small, parabolic.		
Characters			
Remarks	This species is the type species of Balcoracania.		



Holotype
AUGDF16663
(Australia)
(left: external mould, Whittington et al., 1997; right: rubber cast of the external mould, Paterson et al., 2006)



Reconstruction
(Paterson et al., 2006)



SAMP14560, SAMP14568 Australia;
Paterson et al., 2006



SAMP14556

Emuella dalgarnoi

Order/Family				Order Redlichiida / Suborder Redlichiina / Superfamily Emuelloidea / Family Emuellidae	
Genus		Genus <i>Emuella</i> Pocock, 1970		Author	Pocock K. J.
Etymology		Emu Bay, on Kangaroo Island, Australia			
Species		<i>E. dalgarnoi</i> Pocock, 1970		Author	Pocock K. J.
Etymology		Dalgarno — named for Mr R. C. Dalgarno, co-discoverer of the Flinders Range			
Age	Cambrian, Series 2, Stage 3			Main distribution	Australia
Synonyms /					
Diagnosis		Cephalon subpentagonal; cranidium nearly square; glabella constricted at midlength with an expanded frontal lobe, three pairs of transversely broad glabellar furrows; occipital ring with a small node; palpebral lobe nearly as long as the eye ridge, both stout; fixed cheek palpebral area nearly as wide as the free cheek; genal spine of moderate length. Thorax currently recorded with 58 segments (6+52), divided into prothorax and opisthothorax by the macropleural spine of segment 6. Pygidium extremely small.			
Characters					
Remarks		In frontal lobe of the glabella and in the extension angle of the palpebral lobe and eye ridge it resembles <i>B. dailyi</i> , while in cephalic shape it resembles <i>E. polymera</i> .			
 Holotype AUGDF16659 (Australia; Paterson et al., 2006)  AUGDF16660 (Australia; Pocock 1970)					
 					
Figured specimens AUGDF16662 and AUGDF16660 (both rubber casts of external moulds) (Australia; Paterson et al., 2006)					

Emuella polymera

Order/Family	Order Redlichiida / Suborder Redlichiina / Superfamily Emuelloidea / Family Emuellidae		
Genus	Genus Emuella Pocock, 1970	Author	Pocock K. J.
	Etymology Emu Bay, on Kangaroo Island, Australia		
Species	<i>E. polymera</i> Pocock, 1970	Author	Pocock K. J.
	Etymology polymera, Greek, many, referring to the numerous thoracic segments		
Age	Cambrian, Series 2, Stage 3	Main distribution	Australia
Synonyms	/		
Diagnosis	Cephalon subpentagonal; cranium subsquare; glabella constricted at midlength with a slightly expanded frontal lobe, three pairs of transverse glabellar furrows; palpebral lobe shorter than the eye ridge, both stout; fixed cheek broad; free cheek narrow, genal spine of moderate length. Thorax currently recorded with 48 segments (6+42), divided into prothorax and opisthothorax by the macropleural spine of segment 6. Pygidium extremely small.		
Characters			
Remarks	This species is the type species of Emuella.		



Holotype



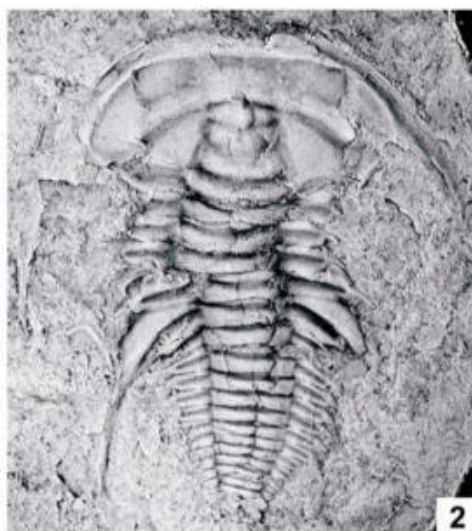
1

(left: Pocock 1970; right: Paterson et al., 2006; both claimed to be internal moulds)



AUGDF16643

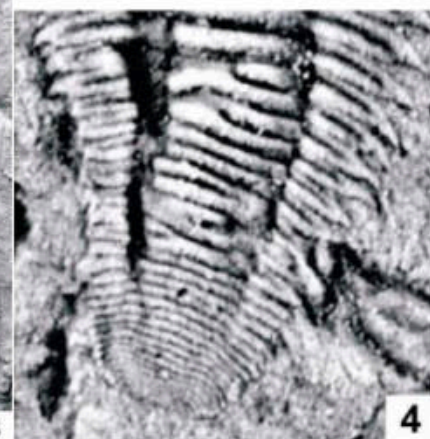
(Australia; Pocock 1970)



2



3



4

Figured specimens (2: rubber cast of external mould, D. Simpson collection; 3, 4: paratypes AUGDF16654)

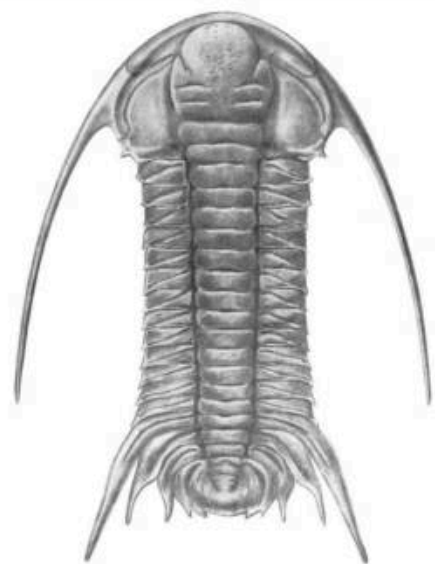
(Australia; Paterson et al., 2006)

Centropleura phoenix

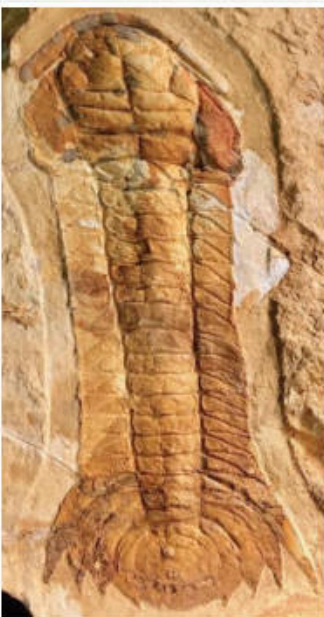
Order/Family	Order Redlichiida / Suborder Redlichiina / Superfamily Paradoxidoidea / Family Centropleuridae		
Genus	Genus Centropleura Angelin, 1854		Author Angelin N. P.
	Etymology centr-, Latin, centre; pleur-, Greek, side, rib		
Species	C. phoenix Angelin, 1854		Author Angelin N. P.
	Etymology phoenix, the mythical bird; the holotype was once burned, symbolising rebirth from the flames		
Age	Cambrian, Miaolingian, Guzhangian	Main distribution	Australia
Synonyms	Centropleura sonax; Centropleura rotundata; Centropleura cf. angelini Centropleura keguqinensis; Centropleura tianshanensis		
Diagnosis	Dorsal shield elongate, hourglass-like. Cephalon semicircular; glabella large, pear-shaped with a prominent front; four pairs of glabellar furrows; palpebral area of the fixed cheek broad; palpebral lobe long; free cheek narrow, genal spine slender and long. Tho Thorax of 16 segments; Anterior 13 segments short pleural segments, almost without pleural spines; Posterior 3 segments with relatively long pleural spines. Pygidium small, broadly oval; Pygidial axis of three rings plus a terminal segment; With two pairs of short marginal spines.		
Characters			
Remarks	/		



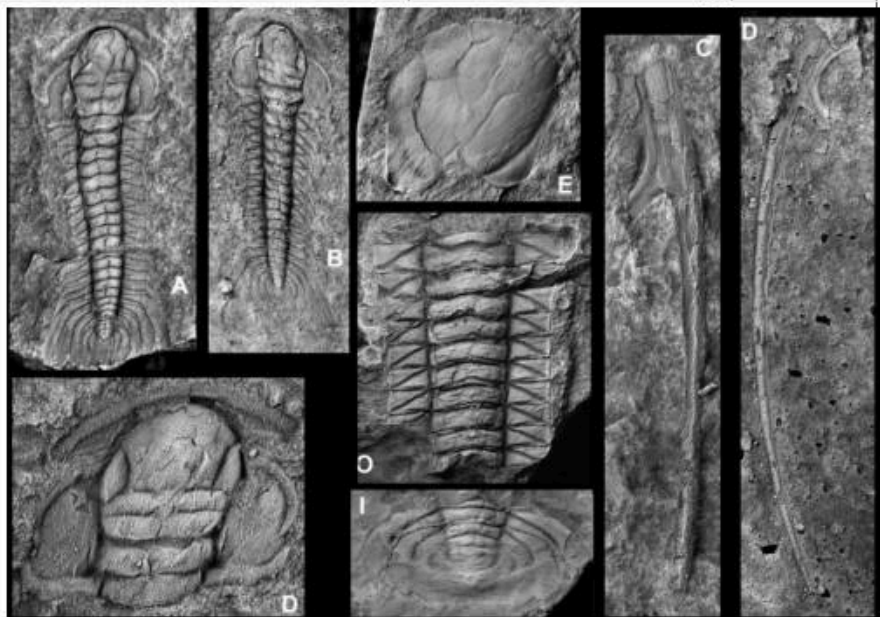
Holotype



Reconstruction (Öpik 1961)



Long 12.4cm Australia; Secher 2024



Figured specimen (Australia; Bentley 2014)

Abadiella huoi

Order/Family Order Redlichiida / Suborder Redlichiina / Superfamily Redlichioidea / Family Abadiellidae			
Genus Genus Abadiella Hupé, 1953		Author	Hup é P.
Etymology Unknown			
Species A. huoi (Chang, 1966)		Author	Zhang Wentang
Etymology Huo — named for the paleontologist Mr Huo Shicheng			
Age	Cambrian, Series 2, Stage 3	Main distribution	Shaanxi, Yunnan, Sichuan; Australia
Synonyms	Parabadiella huoi; Parabadiella hanzhongensis; Parabadiella nanzhengensis Parabadiella(Danangouia)transversa; Parabadiella(Danangouia)quadrata; Parabadiella obscurata Parabadiella shaanxiensis; Parabadiella tenuilimbata; Guangyuanaspis modaoyaensis; Parabadiella longmenshanensis; Guangyuanaspis elegans; Guangyuanaspis shaanxiensis		
Diagnosis Characters	Glabella conical, three pairs of glabellar furrows; Occipital ring with occipital spine; Eye of moderate size, roughly parallel to the axis, posterior end far from the glabella; Eye ridge long; Ocular area of fixed cheek broad; Anterior border short, axial area which projects backward as a plectrum-like extension joining the front of the glabella; genal field of the free cheek broad, genal spine of moderate length. Thorax of 17 segments, axis nearly as wide as the pleural regions; pleural furrows broad and deep, with short pleural spines. Pygidium extremely small, without a border.		
Remarks	This is the earliest trilobite known from China, and one of the earliest trilobites in the world.		



Type specimen (Shaanxi; Whittington et al., 1997)



A



B

Figured specimen (Shaanxi; Wang et al., 2022)



A



B



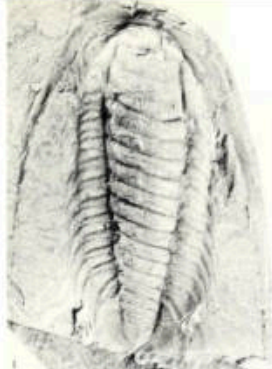
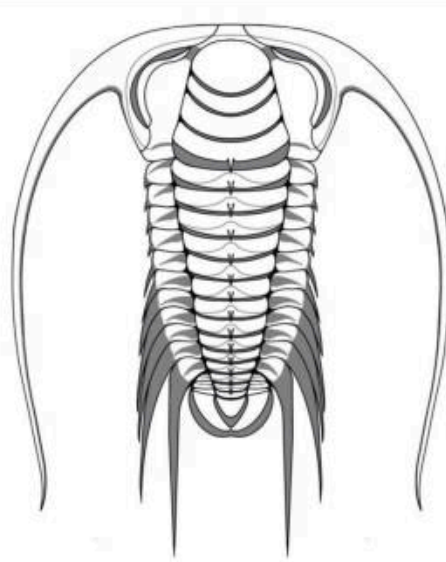
C

Figured specimen (Shaanxi; Zhang et al., 2017)

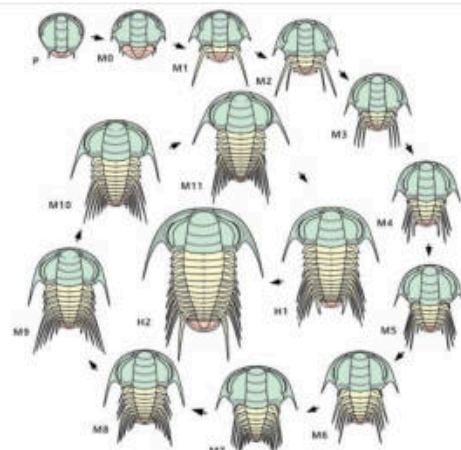
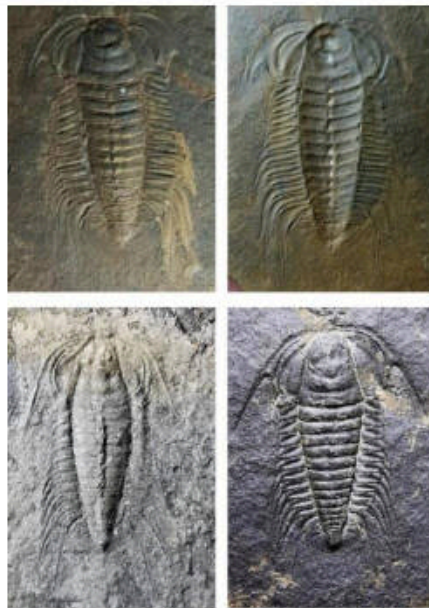
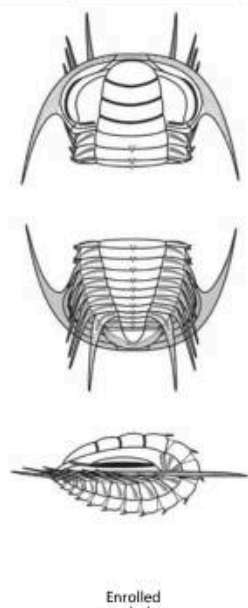
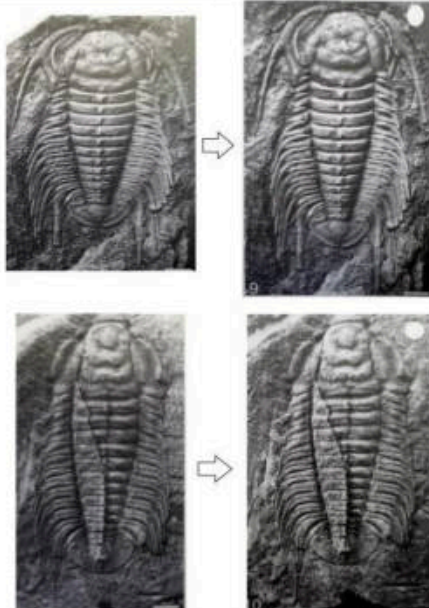


Figured specimen (Shaanxi; Peng Shanchi 2020)

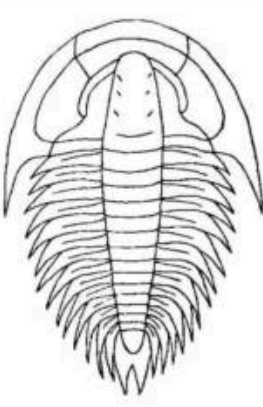
Bathynotus holopygus

Order/Family				Order Redlichiida / Suborder Redlichiina / Superfamily Redlichioidea / Family Bathynotidae	
Genus		Genus Bathynotus Hall, 1860		Author	Hall J.
Etymology		bathy-, Greek, deep, high; not-, Greek, back			
Species		B. holopygus (Hall, 1859)		Author	Hall J.
Etymology		holo-, Greek, whole; pyg-, Greek, tail			
Age	Cambrian, Series 2, Stage 4			Main distribution	Guizhou; North America
Synonyms	Peltura(Olenus)holopyga; Bathynotus holopyga; B. sinensis				
Diagnosis	Dorsal shield elongate-oval. Cranidium sub-trapezoidal, gently convex; glabella broadly conical, rounded anteriorly, with two pairs of transverse glabellar furrows slightly arched backward; inner border absent or extremely narrow; occipital furrow clearly defined occipital ring of uniform width, bearing small tubercles; palpebral lobe slender and ridge-like, palpebral furrow broad and deep; fixed and free cheeks of equal width, both rather narrow; free cheek with a huge, long genal spine directed backward and Protruding below; Hypostome distinctive, nearly pentagonal. Thorax with thirteen segments; Axis twice as wide as the pleural region, with small tubercles; Pleural region narrow, 11th segment with longest pleural spine, 12th-13th segments with pleural spines not developed. Pygidium micropygous, elongate-oval and transversely broad; axis relatively broad, usually with one narrow axial ring and a large terminal piece; border broad and flat.				
Characters					
Remarks	This species is the type species of Bathynotus.				
 Early illustrated specimens Walcott 1886 (USA; cited in Whittington 1988)  Reconstruction (dorsal view) by Zhuo et al. 2020  14-1 (Guizhou; Zhao Xiaoyun et al. 2020)					
 Figured specimen (Guizhou; Zhao Xiaoyun et al., 2020)  Complete dorsal shield (Guizhou; Miaoling Nature School) 					

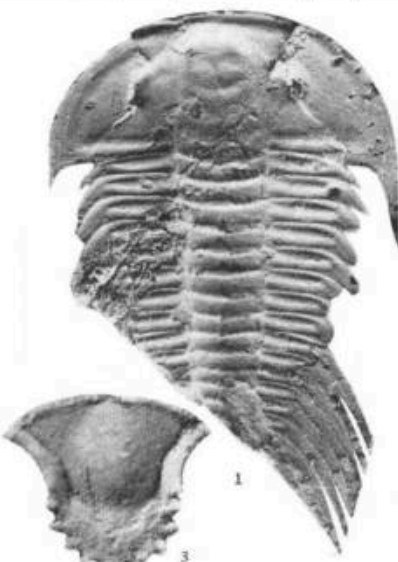
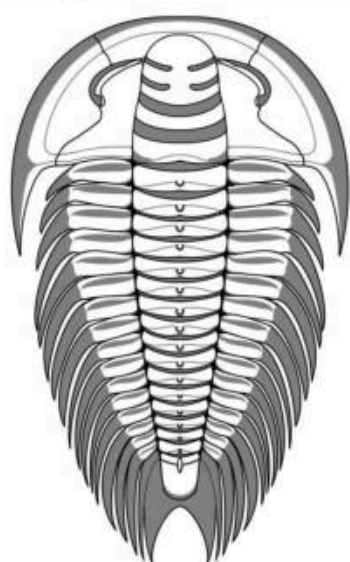
Bathynotus kueichouensis

Order/Family	Order Redlichiida / Suborder Redlichiina / Superfamily Redlichiioidea / Family Bathynotidae		
Genus	Genus Bathynotus Hall, 1860	Author	Hall J.
Etymology	bathy-, Greek, deep, high; not-, Greek, back		
Species	B. kueichouensis Lu in Wang, 1964	Author	Lu Yanhao
Etymology	Kueichou; type specimen type locality		
Age	Cambrian, Series 2, Stage 4	Main distribution	Guizhou, Hunan, Hubei, Anhui; Siberia, North America, Australia
Synonyms	Bathynotus nanjiangensis; B. hunanensis; Bathynotus hubeiensis; Bathynotus gaotanensis Bathynotus elongatus		
Diagnosis	Body elongate-oval. Glabella large, tapering forward, anterior end rounded, with two pairs of long glabellar furrows; Eye long, extending from anterolateral side of glabella backward-outward, turning inward and reaching the posterior border furrow; Border Narrow and short, depressed; Fixed cheek about half as wide as the middle of the glabella; Free cheek narrow, genal spines of moderate length. Thorax of 13 segments; Axis very broad, narrowing backward, axial rings with a median node; Pleural region narrower in front, broader behind, pleural spines gradually lengthening from front to back, 12th and 13th pleural segments small without spines, 11th segment very large. Pygidium semicircular, axis about two-thirds the pygidial length, divided rings indistinct, pleurae slightly wider than the axis, unsegmented or only faintly segmented.		
Characters			
Remarks	Image-based distortion restoration of the type specimen of B. elongatus yields a shape identical to B. kueichouensis, proving it to be a junior synonym of the latter (Peng Shanchi 2020).		
 Type specimen (Guizhou; Peng Shanchi 2020)  Reconstruction (Yang et al., 2021)  Ontogenetic sequence Zhang et al., 2022			
 Complete dorsal shield (Guizhou; Guodong)  Enrolled morphology reconstruction (Yang et al., 2021)  Reconstruction of deformed specimen Peng Shanchi 2020			

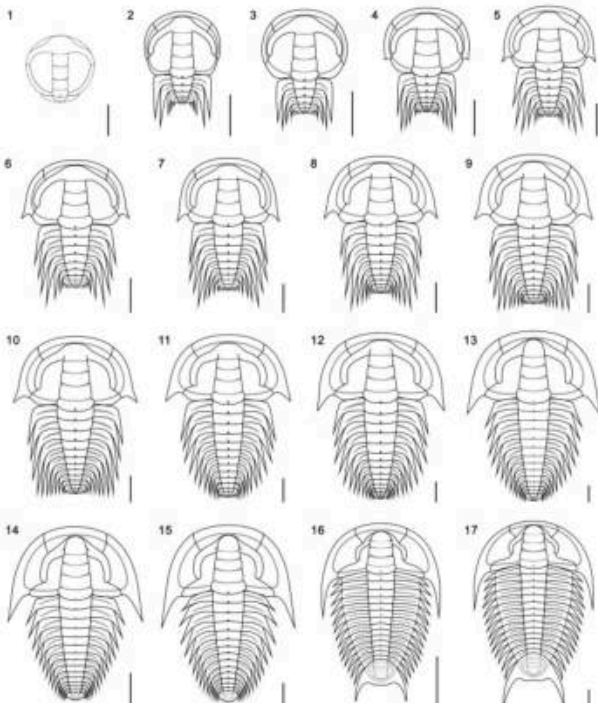
Dolerolenus laevigata

Order/Family	Order Redlichiida / Suborder Redlichiina / Superfamily Redlichioidea / Family Dolerolenidae			
Genus	Genus Dolerolenus Leanza, 1949	Author	Leanza A. F.	
	Etymologydoler, deceptive; Olenus, trilobitegenus name, olenellid			
Species	D. laevigata (Lu, 1961)	Author	Lu Yanhao	
	Etymologylaevigata, Latin, smooth.			
Age	Cambrian, Series 2, Stage 3	Main distribution	Yunnan	
Synonyms	Malungia laevigata			
Diagnosis	Dorsal shield elongate-oval. Cephalon semicircular; glabella conical, with three pairs of weak glabellar furrows; occipital furrow shallow; inner and outer borders of nearly equal width, inner border slightly concave, outer border gently convex; palpebral lobe of moderate size, eye ridge rather weak; fixed cheek narrow, posterior limb triangular; anterior facial suture branches diverging outward, posterior branches directed backward and outward; free cheek broad and flat, genal spine stout. Thorax with 14 segments; pleural regions similar in width to the axis; pleural spines rather long. Pygidium small; axis convex, short and broad; pleural lobes narrow and flat; bearing one pair of scissor-like pygidial spines.			
Characters				
Remarks	/			
				
	Holotype (Yunnan; Lu Yanhao et al. 1965)	Reconstruction (Zhang Wentang 1966)	Figured specimen (Yunnan; Peng Shanchi 2020)	Pathological specimen Yunnan; Templar1st
				
	Complete dorsal shield (Yunnan; Heiyanquan Xiansheng)	Complete dorsal shield, part and counterpart (Yunnan; Pengcheng Jianke)		Complete dorsal shield (Yunnan; Pengcheng Jianke)
				
	Complete dorsal shield (Yunnan; Xuanling)	Complete dorsal shield (Yunnan; Guodong)		

Yiliangella forficula

Order/Family Order Redlichiida / Suborder Redlichiina / Superfamily Redlichioidea / Family Gigantopygidae			
Genus Genus Yiliangella Chang, 1966		Author	Zhang Wentang
Etymology Yiliang; -ella			
Species Y. forficula Chang, 1966		Author	Zhang Wentang
Etymology forficula, Latin, scissors-shaped, referring to the form of the pygidium			
Age	Cambrian, Series 2, Stage 3	Main distribution	Yunnan, Sichuan, Shaanxi
Synonyms	Yiliangella leshanensis; Yiliangella longa		
Diagnosis Characters	Dorsal shield elongate-oval. Cephalon semicircular; cranidium sub-trapezoidal; glabella conical, convex, rounded anteriorly, with three pairs of shallow, broad glabellar furrows; occipital furrow shallow and broad, occipital ring with a node; opistho parian sutures, anterior branches short, posterior branches long; fixed cheek rather narrow, posterior limb broad and long; eye ridge obliquely directed, palpebral lobe rather small; anterior border furrow shallow, inner border absent or extremely narrow; free cheek broad, genal spine rather long. Thorax of 16 segments, each axial ring bearing a small node, pleural spines lengthening from front to back. Pygidium small, axis convex and cylindrical, divided into two segments, with two pairs of long pleural spines extending backward from two pairs of ribs, the two outermost ones large and long.		
Remarks	This species is the type species of Yiliangella.		
			
Type specimen (1 is the holotype; Yunnan; Zhang Wentang 1966)		Reconstruction (drawn by Zhuoyi)	
			
Complete dorsal shield (Yunnan; Chenobei Baigong)		Complete dorsal shield, part and counterpart (Yunnan; Pengcheng Jianke)	
			

Zhangshania typica

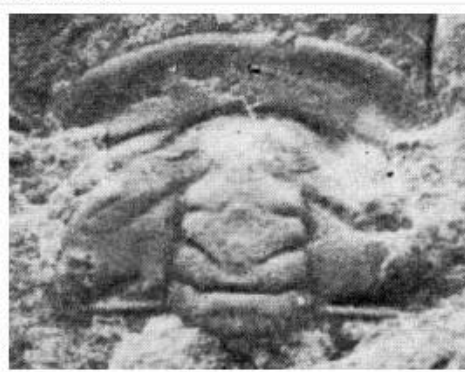
Order/Family	Order Redlichiida / Suborder Redlichiina / Superfamily Redlichioidea / Family Gigantopygidae		
Genus	Genus Zhangshania Li & Zhang in Li et al., 1990	Author	Li Shangi, Zhang Xiguang
	Etymology	Zhangshan, at Fandian Township, Shawan District, Leshan, Sichuan, the type locality	
Species	<i>Z. typica</i> Li & Zhang in Li et al., 1990	Author	Li Shangi, Zhang Xiguang
	Etymology	typica, standard, typical	
Age	Cambrian, Series 2, Stage 3	Main distribution	Yunnan, Sichuan
Synonyms	Parazhangshania sichuanensis		
Diagnosis Characters	Dorsal shield elongate-oval, longer than wide. Cephalon semicircular to semi-elliptical; cranidium sub-trapezoidal, glabella convex, elongate-cylindrical, only slightly tapering forward, with three pairs of furrows connected at midlength; palpebral lobe of moderate size, situated opposite the midlength of the glabella; anterior facial suture branches rather short, meeting the axis at about 40°; posterior branches long, directed outward and backward Extending; Free cheek broad and flat, genal spines relatively long. Thorax long, gradually tapering backward, 14 thoracic segments, each with weak axial node; Pleural lobes flat, pleural spines regularly becoming longer. Pygidial axis convex and elongate-columnar, rounded at the tip; a pair of long spines extends backward on either side.		
Remarks	This species is the type species of Zhangshania.		
 Type specimen s (Sichuan; Li Shanji et al., 1990)  Illustrated specimens Early adult Yunnan; Baishui Ontogenetic sequence reconstruction (Hou et al., 2017) (Yunnan; Hou et al., 2017)   1  5			
 Complete dorsal shield (Yunnan; Yunnan University collection)  Complete dorsal shield (Yunnan; Dai Xu)  Complete dorsal shield (Yunnan; Pengcheng Jianke)  Complete dorsal shield (Yunnan; Zai Lushang)			

Kueichowia liui

Order/Family	Order Redlichiida / Suborder Redlichiina / Superfamily Redlichioidea / Family Kueichowiiidae		
Genus	Genus Kueichowia Lu, 1942	Author	Lu Yanhao
	Etymology Kueichow		
Species	K. liui Lu, 1942	Author	Lu Yanhao
	Etymology Liu — a surname, from Mr Liu Tsing-hsiang of the National Chekiang University, who collected the specimen		
Age	Cambrian, Series 2, Stage 3	Main distribution	Guizhou
Synonyms	Kueichowia xishuiensis; Kueichowia guizhouensis		
Diagnosis Characters	Cranidium nearly square; glabella slightly narrowing forward, broadly rounded anteriorly, with four pairs of glabellar furrows; occipital furrow continuous; preglabellar field long longitudinally, bearing a ridge parallel to the anterior margin of the glabella; eye ridge stout, Extending obliquely backward; Anterior border short and convex longitudinally, anterior border furrow long and shallow longitudinally; Free cheek narrow, genal spines extremely long, extending obliquely outward, intergenal angle large. Thorax with about 14 segments, 4th segment macropleural the first two with dagger-like pleural spines directed outward, later ones longer and directed backward and outward. Pygidium small; axis bearing a pair of rounded prominences.		
Remarks	This species is the type species of Kueichowia.		



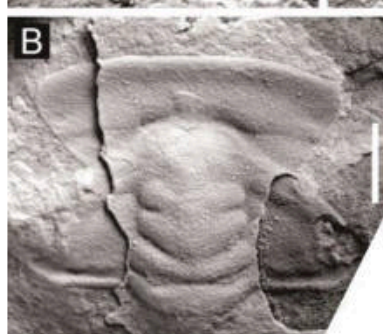
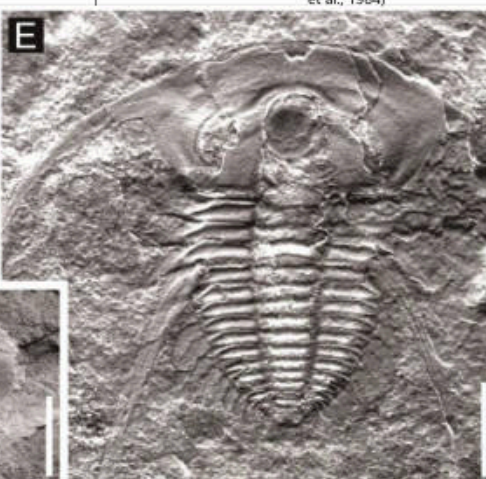
Holotype



Figured specimen (Guizhou; Zhao Yuanlong et al., 1984)



(Guizhou; Sun Zhixin et al. 2025)



(Guizhou; Lixia)

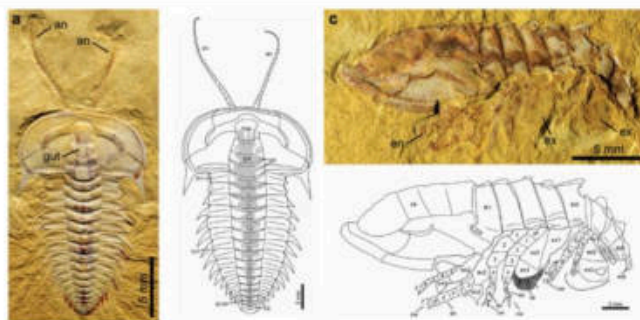
(Guizhou; Sun Zhixin et al. 2025)
(Guizhou; Mr. Y)

Hongshiyanspis yiliangensis

Order/Family	Order Redlichiida / Suborder Redlichiina / Superfamily Redlichioidea / Family Metadoxidae		
Genus	Genus Hongshiyanspis Zhang & Lin in Zhang et al., 1980	Author	Zhang Wentang, Lin Huanling
	Etymology Hongshiya, Hongshiya near Yiliang, Yunnan, the type locality; -aspis, shield, carapace		
Species	H. yiliangensis Zhang & Lin in Zhang et al., 1980	Author	Zhang Wentang, Lin Huanling
	Etymology Yiliang, type specimen type locality		
Age	Cambrian, Series 2, Stage 3	Main distribution	Yunnan
Synonyms	Hongshiyanspis longus		
Diagnosis Characters	Cranidium broad, narrower anteriorly, length to width (transverse width of the outer border) 1:1. Glabella raised, conical, rounded anteriorly, almost reaching the outer border; three pairs of glabellar furrows, the third pair crossing the glabella; occipital furrow deep laterally and directed backward, broad and shallow medially; occipital ring slightly wider medially, with a node; eye ridge raised, long and broad; palpebral lobe short, its posterior end opposite Middle position; Fixed cheek broad, anterior border extending almost horizontally, with a short genal spine at the posterior end; Proparian facial suture, anterior branch short extending forward parallel, posterior branch long. Thorax of 14 segments, the axis gradually narrowing, each segment with an axial node; pleurae broad anteriorly and narrow posteriorly, pleural spines long anteriorly and short posteriorly. Pygidium small, with one fused segment.		
Remarks	This species is the type species of Hongshiyanspis.		



Type specimen (Yunnan; Zhang Wentang et al., 1980)



Specimen preserving antennae and appendages, with reconstruction (Yunnan; Zeng et al., 2017)



Complete dorsal shield (Yunnan; He Ming)



Complete dorsal shield (Yunnan; Xuanling)



Complete dorsal shield (Yunnan; Zai Lushang)
Complete dorsal shield (Yunnan; Zai Lushang)

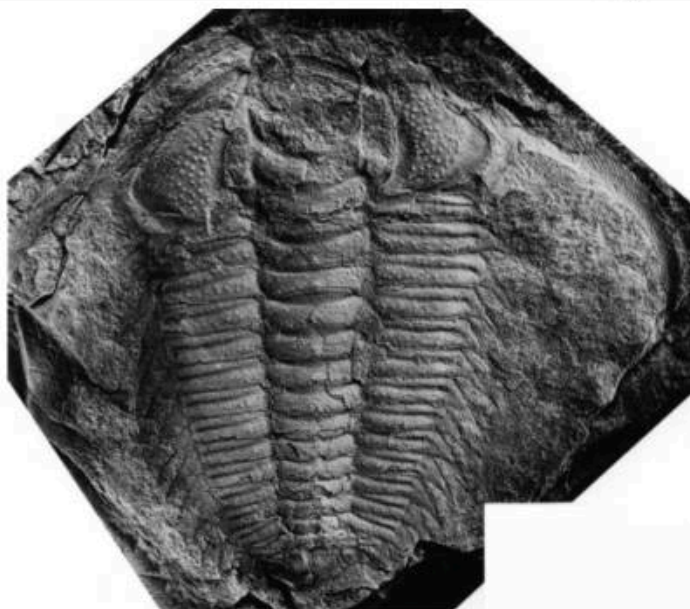


Breviredlichia gaoqiaoensis

Order/Family	Order Redlichiida / Suborder Redlichiina / Superfamily Redlichioidea / Family Redlichiidae		
	Genus Breviredlichia Chang & Lin in Yin & Lee, 1978		Author Zhang Wentang, Lin Huanling
Genus			
	Etymology	brevi-, Latin, short; Redlichia, a trilobite genus name	
		B. gaoqiaoensis Chang & Lin in Yin & Lee, 1978	Author Zhang Wentang, Lin Huanling
Species			
	Etymology	Gaoqiao, Pengshui, Sichuan	
Age	Cambrian, Series 2, Stage 4		Main distribution Sichuan
Synonyms	/		
Diagnosis	Compared with B. granulosa, in this species the glabella does not taper sharply and incline downward in front of the palpebral lobes, its anterior end is broadly rounded rather than pointed, and the frontal lobe is not Shaped swelling.		
Characters			
Remarks	/		



Holotype



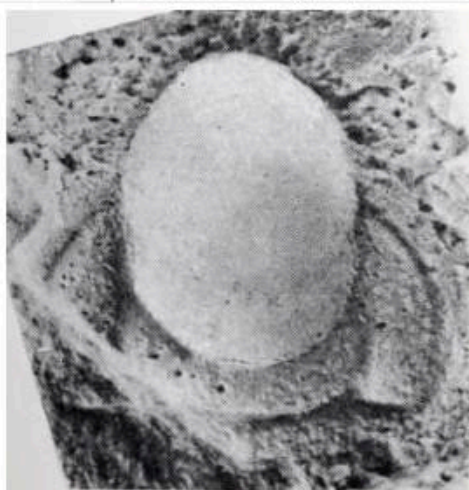
Complete dorsal shield (Sichuan; Whittington et al., 1997)



(Sichuan; Zhang Wentang et al. 1980)

Breviredlichia granulosa

Order/Family	Order Redlichiida / Suborder Redlichiina / Superfamily Redlichioidea / Family Redlichiidae		
	Genus Breviredlichia Chang & Lin in Yin & Lee, 1978		Author Zhang Wentang, Lin Huanling
Genus			
Etymology	brevi-, Latin, short; Redlichia, a trilobite genus name		
	B. granulosa Chang & Lin in Yin & Lee, 1978		Author Zhang Wentang, Lin Huanling
Species			
Etymology	granul-, Latin, small grain, probably referring to the surface of the cranidium and free cheek densely covered with coarse tubercles		
Age	Cambrian, Series 2, Stage 4		Main distribution Yunnan, Guizhou, Sichuan
Synonyms	Breviredlichia shiqianensis		
Diagnosis Characters	Dorsal shield elongate-oval, surface densely tuberculate. Glabella large and broad, barrel-conical, rounded and pointed anteriorly; frontal lobe strongly swollen and bulbous, sloping steeply forward and downward; with two pairs of glabellar furrows, the posterior pair connected medially; occipital furrow deeper and narrower laterally, shallower and broader medially; occipital ring similar to the occipital lobe, with a node; intraocular area narrow, palpebral lobe long, its posterior end reaching the level of the occipital furrow. On, close to the glabella; Preocular area triangular, anterior border narrow and convex, without inner margin; Anterior branch of facial suture short, at about 80 degrees to the cranidial midline; Posterior branch extremely short; Free cheek broad, genal spine stout, intergenal spine present. Thorax of 16 segments, axial rings bearing axial nodes, segment 12 with a short axial spine. Pygidium small and semicircular.		
Remarks	This species is the type species of Breviredlichia.		



Holotype

Complete dorsal shield
(Yunnan; He Ming)Complete dorsal shield
(Yunnan; Guodong)Complete dorsal shield
(Yunnan; Y.W.)

Chengjiangaspis chengjiangensis

Order/Family	Order Redlichiida / Suborder Redlichiina / Superfamily Redlichioidea / Family Redlichiidae		
Genus	Genus Chengjiangaspis Zhang & Lin in Zhang et al., 1980	Author	Zhang Wentang, Lin Huanling
	EtymologyChengjiang; aspis, shield		
Species	C. chengjiangensis Zhang & Lin in Zhang et al., 1980	Author	Zhang Wentang, Lin Huanling
	EtymologyChengjiang		
Age	Cambrian, Series 2, Stage 3	Main distribution	Yunnan
Synonyms	/		
Diagnosis	Dorsal shield elongate-oval. Cranium sub-trapezoidal; glabella truncate-conical; three pairs of glabellar furrows, S1 deep and strongly directed backward, not connected medially; occipital furrow broad and deep; occipital ring broad medially and narrow laterally, Without occipital node; Ocular area of fixed cheek broad; Eye ridge long and stout; Palpebral lobe slightly shorter than eye ridge, posterior end far from the glabella; Free cheek narrow, genal spines short. Thorax of 16 segments; Axis and pleural region subequal in width; pleural furrows broad and deep, pleural spines short. Pygidium small; axis with two rings plus a terminal piece.		
Characters			
Remarks	/		



NIGP37792

(Yunnan; upper: Zhang Wentang et al., 1980; lower: Zhu Yuyan et al., 2024)



Figured specimen (Yunnan; Zhang Wentang et al., 1980)



Complete dorsal shield (Yunnan; He Ming)



Complete dorsal shield (Yunnan; Pengcheng Jianke)





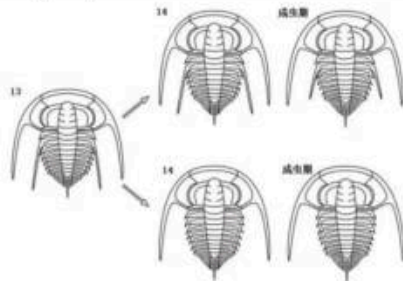
Complete dorsal shield (Yunnan; Xiaozhuang)

Eoredlichia intermedia

Order/Family	Order Redlichiida / Suborder Redlichiina / Superfamily Redlichioidea / Family Redlichiidae		
Genus	Genus Eoredlichia Chang, 1950		Author Zhang Wentang
	Etymology eo-, Greek, dawn, beginning; Redlichia, a trilobite genus name		
Species	<i>E. intermedia</i> (Lu, 1940)		Author Lu Yanhao
	Etymology intermedia, Latin, intermediate		
Age	Cambrian, Series 2, Stage 3		Main distribution Yunnan
Synonyms	Redlichia intermediata; E. (Eoredlichia) intermedia		
Diagnosis Characters	Dorsal shield elongate-oval. Cephalon subsemicircular; glabella conical, broadly rounded anteriorly; three pairs of glabellar furrows, S3 shallow and indistinct, S1 deeper and longer; dorsal furrow distinct; occipital furrow shallow and broad medially, deep and oblique laterally; occipital ring widest medially; palpebral lobe convex, crescentic, its posterior end set well away from the glabella; eye ridge stout, shorter than the palpebral lobe; preocular wing subtriangular; anterior facial suture branches making an angle of Angle with the axis 30–40 degrees; Free cheek broad, genal spines slender and long. Thorax of 15 segments, 9th segment with a stout, long axial spine; Pleural furrow broad and shallow, pleural spines short. Pygidium very small, with a fused segment; with one to two pairs of transverse furrows or pits not connected medially.		
Remarks	This species is the type species of Eoredlichia.		



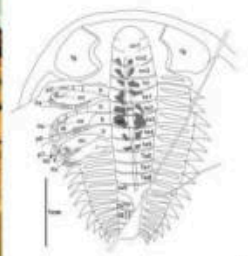
Type specimen
(Yunnan; Peng Shanchi 2020)



Reconstruction (Dai et al., 2013)



Specimen with appendages and digestive glands preserved (Hou et al., 2009)



Specimens of various developmental stages (Yunnan; Dai et al., 2023)



Complete dorsal shield
(Yunnan; Acuan)



Kepingaspis kepingensis

Order/Family	Order Redlichiida / Suborder Redlichiina / Superfamily Redlichioidea / Family Redlichiidae		
Genus	Genus Kepingaspis Chang, 1965	Author	Zhang Tairong
Etymology	Keping, pinyin, the type locality		
Species	K. kepingensis Chang, 1965	Author	Zhang Tairong
Etymology	Keping, pinyin, the type locality in Keping, Xinjiang		
Age	Cambrian, Series 2, Stage 3	Main distribution	Xinjiang
Synonyms	/		
Diagnosis Characters	Dorsal shield narrowly elongate-oval, gently convex. Cephalon semicircular; cranidium subsquare; glabella conical, gently convex, rounded anteriorly; three pairs of glabellar furrows directed obliquely backward; occipital furrow directed obliquely forward laterally; occipital ring with a short occipital spine; palpebral lobe long and hook-shaped; eye ridge slightly shorter than the palpebral lobe, its front end reaching the first glabellar furrow; fixed cheek narrow; free cheek broad, genal spine small; facial Anterior branch of facial suture slightly obliquely outward then cutting inward to the margin; Posterior branch obliquely backward and outward. Thorax relatively narrow; 19 Segments, each axial ring with small median tubercle; Pleural region about as wide as the axis, pleural furrow Wide, pleural spines short. Pygidium very small, subcircular, divided into two segments.		
Remarks	This species is the type species of Kepingaspis.		



Type specimen (Xinjiang; Zhang Tairong 1965)



Complete dorsal shield (Xinjiang; Guodong)



Complete dorsal shield (Xinjiang; Baishui)

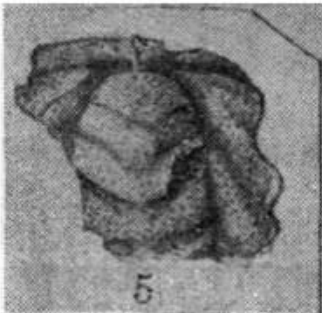






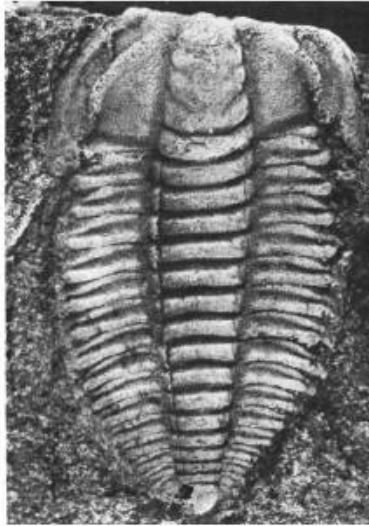
Complete dorsal shield (Xinjiang; Xiaoyu)

Kuanyangia pustulosa

Order/Family	Order Redlichiida / Suborder Redlichiina / Superfamily Redlichioidea / Family Redlichiidae		
Genus	Genus Kuanyangia Hupé, 1953	Author	Hup é P.
Etymology	Kuanyang, a compound of the surnames Kuan and Yang in Wade-Giles romanisation, from the collectors S. T. Kuan and T. H. Yang		
Species	K. pustulosa (Lu, 1941)	Author	Lu Yanhao
Etymology	pustul-, Latin, pustule, node, referring to the cephalon surface covered with pustule-like protuberances		
Age	Cambrian, Series 2, Stage 3	Main distribution	Yunnan
Synonyms	Redlichia pustulosa; Kuanyangia(Kuanyangia)pustulosa		
Diagnosis Characters	Dorsal shield elongate-oval, surface with coarse granulate tubercles. Cranidium sub-trapezoidal; glabella broadly conical, three pairs of obliquely directed glabellar furrows connected medially; dorsal furrow broad and shallow; occipital furrow oblique laterally, Middle part straight; Eye ridge long, palpebral lobe slightly shorter, distal end far from the glabella; Anterior branch of facial suture at about 30 degrees to the midline; Fixed cheek relatively broad; Preocular area and posterior limb long transversely, relatively narrow longitudinally; Anterior border of moderate width, inner margin extremely narrow or absent. Thorax gradually tapering backward; 16 Thoracic segments, axis as wide as pleural region, rings with median nodes; Pleural furrow broad and smooth, pleural spines rather long. Pygidium small, with one fused segment.		
Remarks	This species is the type species of Kuanyangia.		
 Original type specimen (lost; Yunnan; Lu 1941)  Topotype  Figured specimen (Yunnan; Hou et al., 2004)			
 Complete dorsal shield (Yunnan; Xiaozhuang)  Complete dorsal shield (Yunnan; Qingshui Fu Qingyi)  Complete dorsal shield (Yunnan; Pengcheng Jianke)			
    Complete dorsal shield, part and counterpart (Yunnan; Guodong)			

Malongocephalus yunnanensis

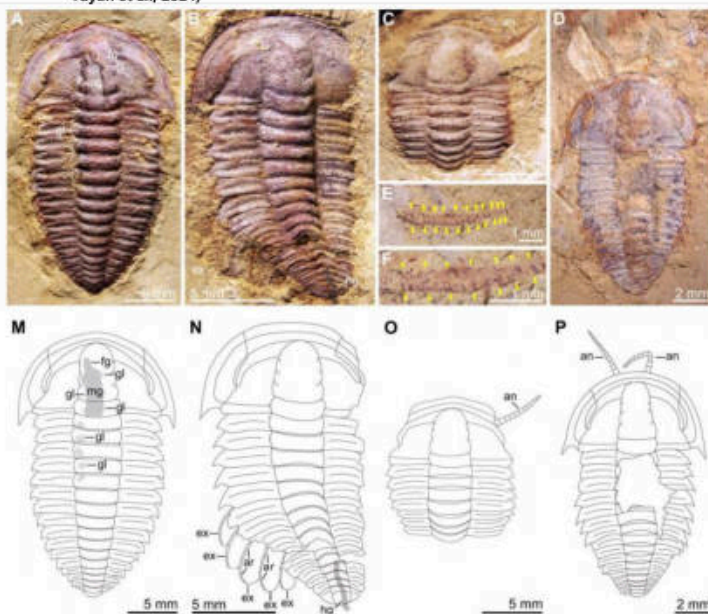
Order/Family	Order Redlichiida / Suborder Redlichiina / Superfamily Redlichioidea / Family Redlichiidae		
Genus	Genus Malongocephalus Zhang & Lin in Zhang et al., 1980	Author	Zhang Wentang, Lin Huanling
	Etymology Malong, Malong, Yunnan; -cephalus, Greek, head		
Species	M. yunnanensis Zhang & Lin in Zhang et al., 1980	Author	Zhang Wentang, Lin Huanling
	Etymology Yunnan		
Age	Cambrian, Series 2, Stage 3	Main distribution	Yunnan
Synonyms	/		
Diagnosis	Dorsal shield elongate-oval. Cephalon semicircular; glabella conical, three pairs of glabellar furrows; preglabellar field of moderate length; palpebral lobe and eye ridge long; free cheek narrow, genal spine short. Thorax of 15 segments; axis as wide as the pleural lobes; pleural furrows broad and deep, pleural spines short and small. Pygidium small and semicircular.		
Characters			
Remarks	This species is the type species of Malongocephalus. (2019) Considered this genus a junior synonym of Chengjiangaspis. Zhu Yuyan et al. (2024) Considered that this species still shows clear differences from Chengjiangaspis chengjiangensis in cranidial details, axis width and number of thoracic segments, they regard them as distinct species, although whether they belong to the same genus is left undecided.		



Holotype NIGP 37826 (Yunnan; left: Zhang Wentang et al., 1980; right: Zhu Yuyan et al., 2024)



Complete dorsal shield (Yunnan; Weidingzhong Guchong)

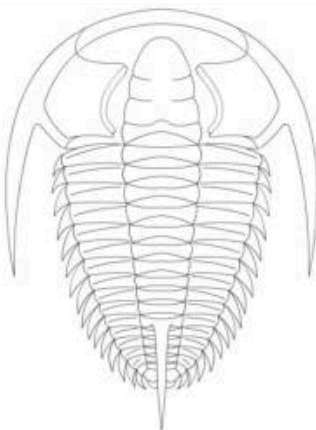


Specimen preserving soft tissues (Yunnan; Zhu Yuyan et al., 2024)



Complete dorsal shield (Yunnan; Luo Yanan)

Redlichia (Pteroredlichia) chinensis

Order/Family	Order Redlichiida / Suborder Redlichiina / Superfamily Redlichioidea / Family Redlichiidae		
Genus	Redlichia	Pteroredlichia	Redlichia(Pteroredlichia)Chang,1966
Etymology	Redlich, a surname, honouring the British paleontologist K. A. Redlich; pter, wing(ed)		Author Zhang Wentang
Species	R. (P.) chinensis Walcott, 1905		Author Walcott C. D.
Etymology	chinensis, Latinisation of Chinese, of China		
Age	Cambrian, Series 2, Stage 4		Main distribution Hunan, Guizhou, Liaoning, Shandong, Anhui, Yunnan
Synonyms	Redlichia chinensis		
Diagnosis	Dorsal shield elongate-oval, cephalon semicircular. Cranidium wider than long; glabella elongate-conical, slightly expanded and rounded anteriorly, with three pairs of transversely directed glabellar furrows; eye ridge short, palpebral lobe long and crescentic, its posterior end curving inward near the occipital furrow; inner border narrow, outer border broad and distinctly convex; anterior facial suture branches extending horizontally, nearly at right angles to the axis; preocular wing wing-like; fixed cheek narrow; free Cheek broad, genal spines long. Thorax of 15 segments, axis broad, tapering backward, slightly broader than the pleural region; Pleural furrow broad and deep, pleural spines relatively short; An axial spine of moderate length developed on the 11th thoracic segment; cheek broad, genal spine long. Pygidium small, with one fused segment, posterior margin concave.		
Characters			
Remarks	/		
 Type specimen (Shandong; Walcott 1913)  Reconstruction (drawn by Haixiang)  Complete dorsal shield (Hunan; Baishui)			
 Complete dorsal shield (Hunan; Baishui)  Complete dorsal shield (Hunan; Baishui)			

Redlichia (Pteroredlichia) chinensis lui

Order/Family	Order Redlichiida / Suborder Redlichiina / Superfamily Redlichioidea / Family Redlichiidae		
Genus	Redlichia Pteroredlichia Redlichia(Pteroredlichia)Chang,1966	Author	Zhang Wentang
Etymology	Redlich, a surname, honouring the British paleontologist K. A. Redlich; pter, wing(ed)		
Species	R. (P.) chinensis lui Chang, 1966	Author	Zhang Wentang
Etymology	chinensis, designation of Chinese, of China; Lu, a surname, probably honouring the distinguished Chinese		
Age	Cambrian, Series 2, Stage 4	Main distribution	Henan, Hunan, Guizhou
Synonyms	/		
Diagnosis	This subspecies is extremely similar to R. (P.) Sinensis, including the glabellar shape and the axial spine developed on the 11th thoracic segment, but its lower end of the palpebral lobe is farther from the dorsal furrow and the anterior branch of the facial suture extends more horizontally extension, a greater transverse length of the preocular wing, a cranial length:width ratio greater than 1:1.5, a thorax of 16 segments, and relatively longer pleural spines.		
Characters			
Remarks	/		



Type specimen (Henan; Peng Shanchi 2020)



Figured specimen (Henan; Peng Shanchi 2020)

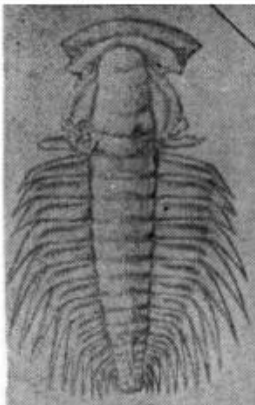


Probable R. (P.) chinensis lui, complete enrolled dorsal shield in nodule (Guizhou; Baishui)

Redlichia (Pteroredlichia) murakamii

Order/Family	Order Redlichiida / Suborder Redlichiina / Superfamily Redlichioidea / Family Redlichiidae		
Genus	Redlichia Pteroredlichia Redlichia(Pteroredlichia)Chang,1966	Author	Zhang Wentang
Etymology	Redlich, a surname, honouring the British paleontologist K. A. Redlich; pter, wing(ed)		
Species	R. (P.) murakamii Resser & Endo in Kobayashi, 1935	Author	Resser C. E. & Endo R.
Etymology	Murakami, a Japanese surname		
Age	Cambrian, Series 2, Stage 4	Main distribution	Liaoning, Hunan, Guizhou, Anhui
Synonyms	Redlichia manchuriensis; Redlichia angulata; Redlichia murakamii		
Diagnosis	Body rather small, dorsal shield subelliptical. Cranium semicircular, glabella slightly pointed-conical, three pairs of glabellar furrows; palpebral lobe slender, crescentic, its posterior end near the occipital ring; anterior facial suture branches straight, Relatively short; Free cheek relatively narrow, genal spines long. Thorax of 15 segments, posterior end contracted into an inverted triangle; Axial spine developed in the 4th segment at meraspid stage, disappearing in adults; 11Th segment with long axial spine; pleural spines gradually lengthening from front to back. Pygidium small.		
Characters			
Remarks			
    Holotype (Liaoning; Zhang et al., 1987) (Liaoning; Zhang et al., 1987) Figured specimen (Liaoning; Kobayashi 1935)			
    Complete dorsal shield (Liaoning; tomorrow)			
    Complete dorsal shield (Liaoning; Baishui) Complete dorsal shield, part and counterpart (Liaoning; Guodong) Complete dorsal shield (Hunan; Guodong)			

Redlichia (Redlichia) mai

Order/Family	Order Redlichiida / Suborder Redlichiina / Superfamily Redlichioidea / Family Redlichiidae		
Genus	Genus Redlichia, subgenus Redlichia Cossmann, 1902	Author	Cossmann M.
Etymology	Redlich, a surname, honouring the British paleontologist K. A. Redlich		
Species	R. (R.) mai Lu, 1941	Author	Lu Yanhao
Etymology	Ma — named for Dr Ma H. Y.		
Age	Cambrian, Series 2, Stage 4	Main distribution	Yunnan
Synonyms	Redlichia mai		
Diagnosis Characters	Dorsal shield oval. Cranidium about as long as wide; glabella columnar, slightly constricted at midlength, three pairs of glabellar furrows; occipital furrow distinct; occipital ring slightly wider medially; outer border broad and convex; inner border distinct; prec Limb triangular; Eye ridge short; Palpebral lobe long, crescentic, posterior end close to the glabella; Posterior limb narrow; Anterior branch of facial suture at nearly 60 degrees to the cranial midline; Free cheek broader than inner fixed cheek, Genal spines long and stout. Thorax subtriangular, 15 segments, gradually narrowing backward; Axis and pleural lobes similar in width, axial rings with small nodes, 11th segment with a stout, long axial spine; ocular wing triangular; eye ridge short; palpebral lobe long, crescentic, its posterior end close to the glabella; posterior limb narrow; anterior facial suture branches at about 60° to the axis; free cheek wider than the intraocular cheek, genal spine long and stout. Thorax subtriangular, 15 segments, narrowing backward; axis and pleural lobes of similar width, axial rings with small nodes, segment 11 with a stout long axial spine; pleural lobes broad and flat, pleural furrows broad and deep, pleural spines rather long. Pygidium extremely small, with one fused segment and a pair of dome-like prominences.		
Remarks	/		
 Line drawing of type specimen (Yunnan; Lu 1941)  Photograph of type specimen (Yunnan; Peng Shanchi 2020)  Reconstruction (drawn by Qingshuifu Qingyi, in Hu Shixue et al., 2013)			
 Complete dorsal shield (Yunnan; xianfengcom)  Complete dorsal shield (Yunnan; Zai Lushang)  Complete dorsal shield (Yunnan; Chengbei Baigong)  Complete dorsal shield (Yunnan; Xuanling)			
 Complete dorsal shield (Yunnan; Guodong)  Complete dorsal shield (Yunnan; Guodong)  Complete dorsal shield (Yunnan; Guodong)			